

FullStack

1. Web Foundations and HTTP Basics

- **Documentation:**
 - **MDN Web Docs:** The ultimate resource for HTML, CSS, and JavaScript foundations. [MDN](#)
 - **HTTP MDN Guide:** Learn headers, methods, status codes, and caching. [MDN](#)
- **YouTube / Video Resource:**
 - **SuperSimpleDev:** 6-hour HTML & CSS Full Course (perfect for absolute beginners). [HTML & CSS Full Course - Beginner to Pro](#)
 - **Traversy Media:** Crash courses on HTML, CSS, and HTTP networking basics. <https://youtu.be/UB1O30fR-EE?si=3kf4i49ydV62CYSi>
- **Hands-on:**
 - **freeCodeCamp:** Responsive Web Design Certification. <https://www.freecodecamp.org/learn/responsive-web-design-v9/>
 - **Frontend Mentor:** Real-world challenges to build layouts from professional Figma designs. <https://www.frontendmentor.io/challenges>

2. Git, GitHub, and Project Tooling

- **Documentation:**
 - **Pro Git Book:** The official, comprehensive book available for free on the Git website. <https://git-scm.com/book/en/v2>
 - **GitHub Docs:** Guides on pull requests, branching, and merge conflicts. <https://docs.github.com/en/pull-requests>
- **YouTube / Video Resource:**
 - **Kunwar Kushwaha:** Complete Git and GitHub Tutorial. <https://youtu.be/apGV9Kg7ics?si=WaQbjZyWelj6JSx0>
 - **Amigoscode:** Git and GitHub crash course for beginners. https://youtu.be/3fUbBnN_H2c?si=Dkqx1gQmx4_fIUrq
- **Hands-on:**
 - **Learn Git Branching:** An interactive visual sandbox simulator for practicing Git commands. <https://learngitbranching.js.org/>
 - **GitHub Minesweeper / First Contributions:** Open-source repositories designed for making your first pull request. <https://github.com/firstcontributions/first-contributions>

3. React, Tailwind CSS, and Component UI

- **Documentation:**
 - **React.dev:** The updated official docs featuring interactive code sandboxes. <https://react.dev/learn>
 - **Tailwind CSS Docs:** Extremely intuitive search and copy-paste design utility docs. <https://tailwindcss.com/>
- **YouTube / Video Resource:**
 - **Programming with Mosh:** React JS Tutorial for Beginners. <https://youtu.be/SqcY0GIETPk?si=wNyjZ9Pnalv1oc3M>
 - **ByteGrad / Dave Gray:** Comprehensive modern React guides with TypeScript. <https://youtu.be/SqcY0GIETPk?si=pqBDaxxgaCutiMmU>
- **Hands-on:**
 - **Scrimba:** Learn React course (allows you to pause and edit code directly inside the video player). <https://codesandbox.io/p/sandbox/>
 - **Build projects:** Create a personal portfolio or a dashboard using Tailwind UI elements. <https://scrimba.com/learn-react-c0e>

4. Backend Foundations with Node.js

- **Documentation:**
 - **Node.js Official Documentation:** Guides on the event loop, file system (`fs`), and asynchronous programming. <https://nodejs.org/api/fs.html>
- **YouTube / Video Resource:**
 - **Dave Gray:** Node.js Full Course for Beginners. <https://youtu.be/f2EqECITBL8?si=0kiaPmwFOPg3yAgm>

- **FreeCodeCamp:** Node.js and Express.js tutorial by John Smilga. https://youtu.be/Oe421EPjeBE?si=SIPsCqxSZ_Ghl0-a
- **Hands-on:**
 - **Nodeschool.io:** Text-based, terminal-driven interactive workshops. <https://nodeschool.io/building-workshops.html>
 - **Exercism:** Node.js track featuring mentor-reviewed coding challenges. <https://exercism.org/>

5. RESTful APIs and CRUD Applications

- **Documentation:**
 - **Express.js Docs:** The standard routing and middleware reference for Node. https://developer.mozilla.org/en-US/docs/Learn_web_development
 - **MDN REST API Guide:** Core architectural principles of REST. <https://developer.mozilla.org/en-US/docs/Glossary/REST>
- **YouTube / Video Resource:**
 - **Academind:** Building a REST API with Node.js. https://youtu.be/bIQ60skPzI0?si=TXVB-9ggyn_b-rf8
 - **Web Dev Simplified:** Real-world REST API design practices. <https://youtu.be/etKM5-gGwto?si=Jq7dvEb69WnKFI7g>
- **Hands-on:**
 - **Postman Academy:** Interactive training to test and build APIs using Postman. <https://academy.postman.com/>
 - **Build a Todo App / E-commerce Backend:** Standard portfolio projects connecting Express routes to a database. <https://www.geeksforgeeks.org/mern/e-commerce-websites-backend-setup/>

6. Authentication, Redis, and Data Persistence

- **Documentation:**
 - **MongoDB Docs / PostgreSQL Docs:** Depending on whether you choose NoSQL or SQL. <https://www.mongodb.com/resources/basics/databases/nosql-explained/nosql-vs-sql>
 - **Redis University:** Official documentation and free structured courses for caching. <https://redis.io/tutorials/university/>
 - **JWT.io:** Essential for understanding JSON Web Tokens. <https://www.jwt.io/introduction#what-is-json-web-token>
- **YouTube / Video Resource:**
 - **Hussein Nasser:** Deep dives into database engineering and caching strategies. https://youtu.be/ccemOqDrc2I?si=GOBpxHSW_1yeNMF
 - **Ben Awad:** JWT authentication tutorials and secure cookie setups. <https://youtu.be/2PPSXonhIck?si=Na7V7DeZXjRemVWw>
- **Hands-on:**
 - **MongoDB University:** Free hands-on labs with cloud clusters. <https://learn.mongodb.com/>
 - **Hackerrank (SQL Track):** Excellent for practicing raw database queries. <https://www.hackerrank.com/domains/sql>

7. Linux Basics and Cloud Fundamentals

- **Documentation:**
 - **AWS Skill Builder / Google Cloud Skills Boost:** Official introductory cloud paths. https://aws.amazon.com/migration-and-modernization/?trk=25844c96-6128-460e-ba5f-2fc7aa8db616&sc_channel=ps&trk=25844c96-6128-460e-ba5f-2fc7aa8db616&sc_channel=ps&ef_id=Cj0KCQjw79nUBhCgARIsADSHka31iIW1tVdrga8JXTBwFLUv8UIbc-IThkTJzJugVQaIE8FYkViZZt8aAnnnEALw_wcB:G:s&s_kwid=AL!4422!3!808827427667!p!!g!!aws_gcp!23846236484!198027718762&gad_campaignid=23846236484&gclid=Cj0KCQjw79nUBhCgARIsADSHka31iIW1tVdrga8JXTBwFLUv8UIbc-IThkTJzJugVQaIE8FYkViZZt8aAnnnEALw_wcB
 - **The Linux Documentation Project:** Advanced bash scripts and kernel architectures. <https://tldp.org/LDP/abs/abs-guide.pdf>
- **YouTube / Video Resource:**
 - **NetworkChuck:** Free CCNA, Linux, and cloud networking series (high energy, beginner-friendly). <https://youtube.com/playlist?list=PLlhvC56v63IJVXv0GJcl9vO5Z6znCVb1P&si=BnzChZ2EHpFacpLx>
 - **Andrew Brown (ExamPro):** Free 10+ hour cloud certification courses on freeCodeCamp. <https://youtu.be/7HKot-brXFE?si=-XmdqpKO1TVRA3b>

Hands-on:

- **OverTheWire (Bandit):** A gamified terminal challenge to practice SSH and basic Linux commands.

<https://overthewire.org/wargames/bandit/>

- **AWS Free Tier:** Manually spinning up an EC2 instance and configuring security groups.

<https://www.geeksforgeeks.org/devops/amazon-ec2-creating-an-elastic-cloud-compute-instance/>

8. CI/CD, Monitoring, and Delivery Pipelines

• Documentation:

- **GitHub Actions Documentation:** Syntax rules for workflows, actions, and secrets.

<https://docs.github.com/en/actions/reference/workflows-and-actions/workflow-syntax>

- **Docker Docs:** Containers, multi-stage builds, and compose files. <https://docs.docker.com/build/building/multi-stage/>

• YouTube / Video Resource:

- **TechWorld with Nana:** Highly visual breakdowns of Docker and CI/CD automation pipelines.

<https://youtu.be/bhBSInQcQ2k?si=RttLMDIzeGyJijBI>

- **DevOps Directive:** Practical GitHub Actions setup tutorials. <https://youtu.be/Xwpi0ITkL3U?si=YKHnhlioQ7kvBxgl>

• Hands-on:

- **Play with Docker:** A free, browser-based terminal environment to practice clustering and containerization.

<https://www.docker.com/play-with-docker/>

- **Automate your code:** Create an automated GitHub Action that runs a linter and tests on every code push.

<https://docs.github.com/en/actions/tutorials/build-and-test-code>

9. Infrastructure Automation and DevOps Practice

• Documentation:

- **Terraform Registry & Docs:** Standard resource syntax for writing Infrastructure as Code (IaC).

<https://developer.hashicorp.com/terraform/language/syntax/configuration>

- **Ansible Documentation:** Configuration management scripts. <https://www.geeksforgeeks.org/devops/using-ansible-documentation-effectively/>

• YouTube / Video Resource:

- **TechWorld with Nana:** Terraform and Ansible crash courses. <https://youtu.be/bEXfPzoB4RE?si=I8uCYQNoWmgcdAIH>

- **Anton Babenko:** Weekly updates and practical deep-dives into modern Terraform.

<https://www.youtube.com/live/Bo2Cad94dYw?si=POCqpOuManZYix3E>

• Hands-on:

- **HashiCorp Learn:** Official interactive text sandboxes for writing and deploying Terraform scripts.

<https://developer.hashicorp.com/tutorials>

- **Build a VPC:** Write a script that provisions an entire network topology on AWS/GCP automatically.

<https://www.geeksforgeeks.org/cloud-computing/how-to-create-vpc-network-in-gcp/>

10. Production Deployment, Monitoring, and Scaling

• Documentation:

- **Kubernetes.io (K8s Tasks):** The industry standard orchestration documentation. <https://kubernetes.io/docs/home/>

- **Prometheus & Grafana Docs:** Comprehensive systems monitoring setups.

<https://grafana.com/docs/grafana/latest/fundamentals/getting-started/first-dashboards/get-started-grafana-prometheus/>

• YouTube / Video Resource:

- **KubeSimplify (Siamak Sadeghianfar):** Simplified explanations of clusters and nodes. <https://youtube.com/playlist?list=PL5uLNcv9SibDMdNvINsQYRTIOGNXmbHVX&si=cHcQ5n2k8CuumYbU>

- **FreeCodeCamp:** Kubernetes Course for Beginners. https://youtu.be/d6WC5n9G_sM?si=Usbou0mrGvS80zRn

• Hands-on:

- **Killercoda / Play with K8s:** Free interactive browser labs for running live Kubernetes clusters.

<https://killercoda.com/playgrounds/scenario/kubernetes>

- **Grafana Play:** Interactive live dashboards showing real server analytics metrics. Are you currently studying a **specific milestone**, or looking to plan a **timeline** for this entire roadmap? Let me know so I can tailor the best project architectures for your current stage. **AI can make mistakes, so double-check responses Try without personalization**

<https://play.grafana.org/d/to6j8mh/grafana-play-home?from=now-6h&to=now&timezone=utc>